

Term	Description
Activity limitations	The level of functioning an individual may have in executing activities.
Body functions	The physiological functions of body systems (including psychological functions).
Body structures	The anatomical parts of the body such as organs, limbs, and their components.
Casemix	A system that groups patients using information about the patient, their diagnoses, procedures, complexity, and needs. Used for resource allocation.
Causal relationship (coding)	A relationship that exists if a condition is caused by another condition (e.g. on the same death certificate or in a morbidity coding situation where one condition or factor causes another condition).
Classification	An exhaustive set of mutually exclusive categories to aggregate data at a pre-prescribed level of specialisation for a specific purpose.
Cluster	A cluster are postcoordinated entities that are joined using either a forward slash (/) or ampersand (&).
Content Model	A structured framework that captures the knowledge that underpins the definition of an entity.
Derived classification	Classifications, often tailored for use at the national or international level, or for use in a specialty, based upon reference classifications.
Dual coding	The process used in situations where clinical terms are coded in to two different classification systems or versions for purposes of comparison, transition, mapping, casemix grouping or understanding the implications of change from one system to another.
Duration (coding)	The time-period between the onset of the disease and the time admission for care or death.
Environmental factors	The physical, social, and attitudinal environment in which people live and conduct their lives.
Extension code	Extension codes are lists of additional information that can be added to a stem code when users and settings are interested in reporting more detail. Extension codes are not mutually exclusive and should not be used alone in the context of statistical classification but must be added to a stem code. Extension codes may be used alone in another context, e.g. for device documentation. Not all extension codes can be used with every stem code. Extension codes can never appear in the first position in a cluster.

Term	Description
Foundation Component	A large collection of terms and their relationships, which describe health and health-related domains. Underlying data base content that holds all necessary information to generate print versions of the Tabular list and the alphabetical index as well as additional information that is needed to generate specialty linearisations of ICD-11 and country specific modifications. The mention of a term or entity in the foundation exclusively serves ontological purposes and indexing. Mention of a term or entity in the foundation does not mean approval or endorsement of a particular condition.
Impairments	The problems in body function or structure such as a significant deviation or loss.
Integrated coding	Full use of all chapters of ICD-11 (including, codes from Western Medicine and Traditional Medicine (TM1) chapters) for classification of clinical terms.
Modification rule	When coding for mortality from death certificates, the procedure by which an ICD code for the starting point is replaced by another code, due to special instructions.
Precoordination	Stem codes which contain all pertinent information about a documented clinical concept in a pre-combined fashion.
Postcoordination	The use of multiple codes (i.e. stem codes and/or extension codes) together to fully describe a documented clinical concept. Postcoordination is permitted by a matter of principle with all codes, as long as at least one stem code is used.
Primary parent	A higher level entity that covers the full range or scope of another entity.
Reference classifications	Classifications that cover the main parameters of the health system – disease (ICD), disability, functioning, and health (ICF), and health interventions (ICHI).
Secondary parents	The ability to connect a specific entity within the classification to more than one parent.
Sequence (coding)	In the context of mortality coding, a set of conditions reported line by line with a causal relationship between each element.

Term	Description
The shoreline	In the physical world, the shoreline is where a body of water meets the land. For the WHO Family of International Classifications, the “shoreline” is a metaphor that helps to describe the distinction between content in the Foundation and content in Classifications derived from it (“linearizations”). Currently, all ICD entities are included and searchable in the Foundation, but may not be visible in stem codes in classifications like the ICD-11 MMS (Morbidity and Mortality for Statistics). The architecture of the WHO-FIC Foundation allows the extraction of selected entities into a classification (linearization) that can be used for specific use cases and applications. The content and meaning of the entity are unaltered by this extraction. When an entity is added to a classification, it will be added to a code specific for the classification in interest, and we say that the entity is moved “above the shoreline”.
Starting point (coding)	In the context of mortality coding, normally the condition or event reported that started the sequence of acceptable causal relationships ending with the terminal cause of death, or when a death certificate is correctly completed, the condition reported on the lowest used line in Part 1 of the death certificate.
Stem code	Codes in the Tabular list of ICD-11 that can be used alone. Stem codes may be entities or groupings of high relevance in any of the use cases or clinical conditions that should always be described as one single category. Stem codes are designed to ensure that in use cases that require only one code per case, meaningful information is collected.
Terminal cause of death (coding)	The first condition entered on the first used line of Part 1 of the death certificate. Also known as immediate cause of death.
Underlying cause of death	The disease or injury which initiated the train of morbid events leading directly to death, or the circumstances of the unintentional injury or violence which produced the fatal injury.
Verbal autopsy	A method used to ascertain the cause of a death based on an interview with next of kin or other caregivers where no physician can evaluate the deceased.

1 Part 1 - An Introduction to ICD-11

1.1 International Classification of Diseases (ICD)

The International Classification of Diseases and Related Health Problems (ICD) is a tool for recording, reporting and grouping conditions and factors that influence health. It contains categories for diseases and disorders, health related conditions, external causes of illness or death, anatomy, sites, activities, medicines, vaccines and more.

The purpose of the ICD is to allow the systematic recording, analysis, interpretation and comparison of mortality and morbidity data collected in different countries or regions and at different times.

ICD-11 has been designed to serve semantic interoperability of individual data, reusability of recorded data, for use cases other than health statistics, including decision support, resource allocation, reimbursement, guidelines and more.

The role of the WHO in maintenance and implementation support of ICD is described in the WHO constitution. In addition, implementation and use of the most recent revision of ICD in all member states are legally mandated by the WHO International Nomenclature Regulations and reiterated with revisions of ICD adopted by the World Health Assembly. This includes an updating mechanism between revisions and the ongoing development and implementation of the family of diseases- and health-related classifications (WHO-FIC). The ICD as the core classification linked to other related classifications, specialty versions and terminologies.

In health information systems, data needs to be reusable for epidemiological analysis, resource allocation or research, as well as for individual level uses like health documentation, decision support, or reimbursement. ICD provides highly detailed information, far beyond the level of statistical categories, using unique identifiers. This way for example rare diseases, special findings or individual medicines can be recorded and reported.

The ICD is used to translate diagnoses of diseases and other health problems into alphanumeric codes, allowing storage, retrieval, and analysis of the data. The ICD is the international standard diagnostic classification for all general epidemiological and many health management purposes. These purposes include analysis of general health situations in population groups, monitoring of incidence and prevalence of diseases, and examining other health problems in relation to other variables, such as the characteristics and circumstances of the affected individuals. ICD is also suitable for studies of financial aspects of a health system, such as billing or resource allocation.

The ICD has evolved over the past 150 years from an International List of Causes of Death to a comprehensive classification and terminology system. The ontology-based design of ICD-11 and the migration of its sibling classifications ICF and ICHI to the same ontological infrastructure has enabled the full integration of terminology and classification in a common platform. In that way lossless clinical documentation is possible (coding all the necessary details), statistical aggregation is an integrated feature, end-to-end digital solutions are provided and links to other terminologies for other uses are enabled.

The ICD is used to allocate the majority of global health resources. Users of the ICD include physicians, nurses, other health care providers, researchers, health information management professionals, coders, health information technology workers, analysts, policymakers, insurers, patient organisations, and many more.

The ICD is used in various settings at different levels of detail. It therefore includes an information framework that contains a fully specified set of health concepts and their characteristics and relationships from which appropriate code sets can be selected. The ICD11 ensures consistency with traditional use cases of earlier ICD versions, because it has

been built with the past revisions in mind. Past data analyses based on older versions of ICD can be linked to analyses of data based on ICD11.

The ICD can therefore be used to record, classify and use otherwise data recorded under headings such as cause of death, diagnosis, reason for admission, conditions treated, additional diagnoses, risk factors and reason for consultation, which appear on a wide variety of health records and documents from which statistics are derived.

1.1.1 Intended uses

The ICD has been designed to address the needs of a broad range of use cases, such as: causes of death, morbidity, epidemiology, casemix (Diagnosis-Related Groups (DRG)), quality and patient safety, primary care, functioning assessment, research, prevention, substance (medication) or device safety, specific surveillance like antimicrobial resistance (AMR), cancer registration, injury research, as well as ensuring semantic interoperability for clinical documentation, decision support and guidelines or recommendations.

Detailed information and instructions for users on the different use cases are available in other sections of the guide, relating to mortality use and different morbidity uses. Where innovative uses of ICD are envisaged, it is recommended to consult with the WHO to make best use of the flexibility and features of ICD-11.

1.1.2 Classification

A classification is ‘an exhaustive set of mutually exclusive categories to aggregate data at a pre-prescribed level of specialisation for a specific purpose’ (ISO 17115). A classification involves the categorisation of relevant concepts for the purposes of systematic recording or analysis. The categorisation is based on one or more logical rules. The purpose of a health classification varies. For example, it may be used in the analysis of cause of death (mortality), morbidity, or human functioning. Low frequency concepts tend to be grouped but rare concepts may be individually classified.

Coding is the process of assigning a code from a classification to represent a clinical concept for a specific purpose. Coding rules must be incorporated in the classification to achieve consistency of coding and comparability of coded data over time and place. Classifications are complementary to terminologies, since they are designed to be used for standardised coding of information for statistical purposes.

ICD-11 is combining the elements of classification and terminology and is designed to be linked to other terminologies that may provide additional detail or serve different purposes. Coding in ICD-11 can draw on statistical codes and on Uniform Resource Identifiers (URIs).

1.1.3 ICD in the context of the WHO Family of International Classifications (WHO-FIC)

The WHO Family of International Classifications (WHO-FIC) comprises classifications that have been endorsed by the WHO to describe various aspects of health and the health system in a consistent manner.

The WHO-FIC provides standardised building blocks for health information systems and consists of three broad groups: Reference classifications, Derived classifications, and Related classifications.

The Reference and the Derived classifications are based on the Foundation Component, which is a large collection of concepts (with synonyms and preferred terms) and their relationships, which describe health and health-related domains.

Terms and entities related to diseases and health-related problems are organised into the ICD, those pertaining to functioning into the ICF, and those related to interventions into ICHI (International Classification of Health Interventions). Terms from the Foundation Component may be used in more than one Reference classification.

Derived Classifications draw on terms that may come from one or more of the Reference Classifications. Within the WHO-FIC Family, Related classifications are regarded as complementary to the Reference and Derived classifications. Related classifications have their own sets of terms, but can also share terms as part of the WHO-FIC Family.

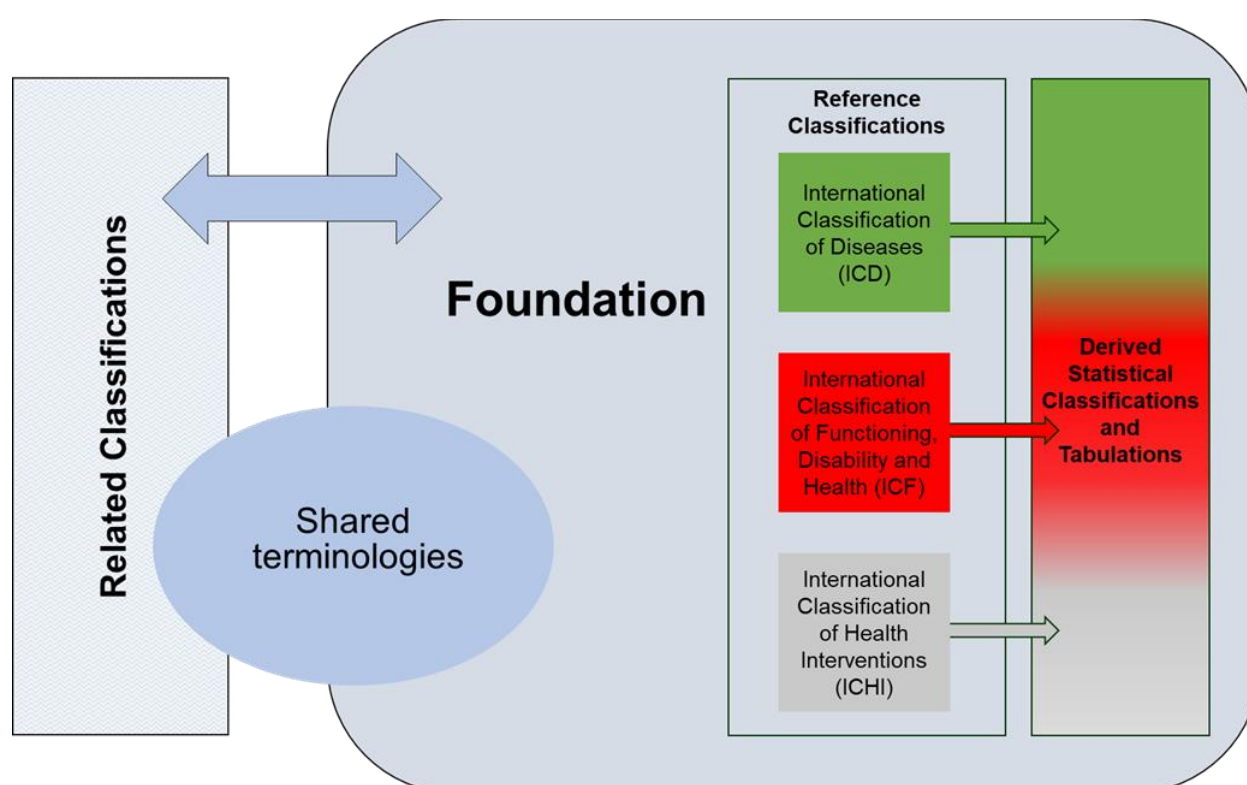


Figure 1: Relationships between the WHO Family of International Classifications (WHO-FIC) and Related Classifications, the Foundation Component, and shared terms.

The purpose of the WHO-FIC is to assist the development of reliable statistical and other data systems at local, national, and international levels, with the aim of improving health status and health care. Health related information might sometimes require additional detail to that contained in the ICD. A group or 'family' of health relevant classifications covers these needs both by classification of domains different from those of the ICD and provision of more details for specific uses, e.g. cancer registration. The WHO-FIC designates a suite of integrated classification products that share similar features and can be used singularly or jointly to provide information on different aspects of health and health care systems. For example, the ICD as a reference classification is mainly used to capture mortality and morbidity information. Functioning is classified in the International

Classification of Functioning, Disability and Health (ICF) and health interventions in the International Classification of Health Interventions (ICHI).

The WHO-FIC provides a conceptual framework of information dimensions which are related to health and health management. In this way, it provides a common language that improves communication and permits comparisons of data within countries, across countries, health care disciplines, services, and time. The WHO and the WHO-FIC Network (including collaborating centres, Non-governmental Organisations (NGOs), and selected experts) ([\[https://www.who.int/standards/classifications/who-fic-maintenance\]](https://www.who.int/standards/classifications/who-fic-maintenance) (<https://www.who.int/standards/classifications/who-fic-maintenance>)) strive to build the Family of International Classifications based on sound scientific and taxonomic principles, ensure that it is up-to-date, culturally appropriate and internationally applicable, and meet the needs of its different users by focusing on the multi-dimensional aspects of health.

1.1.4 WHO-FIC: Reference Classifications

Reference classifications cover the main parameters of the health system, such as death and disease (ICD), disability, functioning, and health (ICF) and health interventions (ICHI). WHO-FIC reference classifications are a product of international agreements. They have achieved broad acceptance and official agreement for use and are approved and recommended as standards for international reporting on health. They may be used as models for the development or revision of other classifications. The three Reference classifications are:

1. International Classification of Diseases and Health Related Problems (ICD)
2. International Classification of Functioning, Disability & Health (ICF)
3. International Classification Health Interventions (ICHI)

The Reference Classifications are based on the same Foundation Component and share sets of Extension Codes.

1.1.4.1 International Classification of Functioning, Disability & Health (ICF)

The ICF is the WHO's framework for measuring health and functioning/disability at both the individual and population levels. While the ICD classifies diseases and causes of death, the ICF classifies health and health-related domains. ICD and ICF together provide a framework to capture the full picture of health.

The ICF classifies health and health-related states in two parts. Part one deals with functioning and disability, described from the perspectives of the body, the individual, and society. It is composed of two components: Body functions and structures, and Activities and participation. Part two covers contextual factors and also has two components: Environmental factors and Personal factors, since an individual's functioning occurs in a context. Work is proceeding on the development of a classification of personal factors to be included in the contextual factors.

Functioning is a generic term for body functions (e.g. memory), body structures (e.g. occipital lobe), and activities and participation (e.g. walking, engaging in paid work). It denotes the positive aspects of the interaction between an individual (related to the individual's health) and that individual's contextual factors (environmental and personal factors).

Disability is an umbrella term for impairments, activity limitations and participation restrictions. It denotes the negative aspects of the interaction between an individual (with a health condition) and that individual's contextual factors (environmental and personal factors). Disabilities are envisioned as a continuum and therefore the ICF and the codes within it do not confer an international binary status of disabled/not disabled. Levels of disability can be used descriptively in clinical settings when formulating a case. Program and policy decision-makers can apply the ICF and specify their own standards for the level of disability as eligibility criteria that are relevant for specific purposes.

ICF includes other relevant descriptions:

- *Body functions* are the physiological functions of body systems (including mental functions).
- *Body structures* are anatomical parts of the body such as organs, limbs and their components.
- *Impairments* are problems in body function or structure such as a significant deviation or loss.
- *Activity* is the execution of a task or action by an individual.
- *Activity limitations* are difficulties an individual may have in executing activities.
- *Participation* is involvement in a life situation.
- *Participation restrictions* are problems an individual may experience in involvement in life situations.
- *Environmental factors* make up the physical, social and attitudinal environment in which people live and conduct their lives.

ICF includes codes for Body Functions (b), Body Structures (s), Activities and Participation (d), and Environmental Factors (e).

ICF codes are only complete with the presence of a qualifier, which denotes the level of health (i.e. severity of the problem from 'no problem' to 'complete problem'). Without qualifiers, codes have no inherent meaning. The ICF acknowledges that every human being can experience a decrement in health and thereby experience some disability. Disabilities can be temporary and may be brief (such as staying home from work for a few days with the flu); they can also be chronic or permanent and may fluctuate in severity over time.

Personal factors are included in the ICF model of functioning and disability, but have not been classified in ICF because of the large social and cultural variants associated with them. The taxonomy has not been operationalized further due to challenges related to establishing cross-cultural applicability, resource constraints, and ethical considerations.

1.1.4.2 The International Classification of Health Interventions (ICHI)

ICHI includes interventions across all functional sectors of the health system, covering acute care, primary care, rehabilitation, assistance with functioning, prevention, public health, and ancillary services. Interventions delivered by all types of providers have been included. The importance of describing and classifying health interventions has long been understood. An International Classification of Procedures in Medicine (ICPM) was published by WHO in 1978 but was not maintained. ICHI is much broader than the former ICPM because it includes the full range of health interventions. Development of ICHI began in 2007, as a joint effort of the

WHO Family of International Classifications (WHO-FIC) Network and WHO, based on the experience gathered with the development of several national classifications of health interventions.

Table 1: *Descriptions and terms used in creation of ICHI.*

Axes	Inclusions	Example
The Target axis contains the entities on which the action is carried out.	Anatomy, Human function, Person or client, Group or population	stomach, activities of daily living
The Action axis is defined as a deed which is done by an actor to a target during a health care intervention.	Investigation, Treating, Managing, Informing, Assisting, Preventing	biopsy, vaccination
The Means axis contains the entities describing the processes and methods by which the action is carried out.	Approach: the process by which the target of the action is accessed	open, endoscopic
	Technique used as part of the action	radiation, magnetic resonance
	Method describing how the action is undertaken	law enforcement, method of transport.

The content of the axes has been restricted to attributes that are common to many interventions. In particular:

- Devices have not been included as an axis because most interventions do not involve a device and devices change rapidly, however devices may be included as extension codes
- Drugs or other substances administered through an intervention are classified elsewhere (e.g. ICD, The Anatomical Therapeutic Chemical Classification with Defined Daily Doses (ATC/DDD), INN).

The coding system comprises a seven-character category structure for the three axes:

- Three letters for the Target
- Two letters for the Action
- Two letters for the Means

ICHI codes comprise valid seven letter combinations of the three axes. For each intervention included in ICHI, the appropriate seven letter combination is identified. Not every possible combination of the three axes represents a valid ICHI domain.

1.1.4.3 WHO-FIC: Derived classifications

Derived classifications are often tailored for use at the national or international level or for use in a particular specialty. They are based on reference classifications (i.e. ICD, ICF, ICHI). Derived classifications may be prepared by:

- adopting the reference classification structure and classes
- providing additional detail beyond that provided by the reference classification
- rearrangement or by aggregation of items from one or more reference classifications.

ICD-11 has specialty linearisations that are derived from the common foundation. These include a version for dermatology, one for primary care and one for mental health. Others may follow.

1.1.4.4 Related classifications

Related classifications are included in the WHO Family of International Classifications to describe important aspects of health or the health system not covered by reference or derived classifications. Related classifications are:

- International Classification of Primary Care (ICPC)
- International Classification of External Causes of Injury (ICECI)
- Technical aids for persons with disabilities (ISO9999)
- The Anatomical Therapeutic Chemical Classification with Defined Daily Doses (ATC/DDD)
- The International Classification for Nursing Practice (ICNP)

1.1.5 ICD use in health information systems

Health information systems include a range of different components for collection, analysis, and use of health data. Information sources could, for example, be population-based, health facility-based, or focused on particular diseases. The main population-based sources of health information are civil registration and vital statistics (CRVS) systems, census data, and household surveys.

Health facility-related data sources include public health surveillance, health services data (that may be referred to as health management information systems or routine health information systems (RHIS)), and health system monitoring data (e.g. human resources, health infrastructure, financing).

National health accounts (NHA) are designed to provide a comprehensive picture of health financing. Coding enables the recording of health information in a language independent way. Standardisation of coding enables both intra- and international data comparison. For example, ICD coded data can be compared across different sectors of the health system – if the same coding rules are applied.

Health information systems are increasingly based on digital (electronic) reporting and coding. ICD-11 is designed to be used in such environments. The content of this Reference Guide is the only additional document required when coding with ICD-11.

In many places information collection is based on paper reporting in a traditional analogue way. ICD-11 can also be produced in a printed version for use in paper-based systems if needed (see [1.1.5.2](#)).

1.1.5.1 Use of ICD–11 in a digital setting and with web services

ICD-11 is used for coding of diagnoses, in electronic health records or with electronic death certificates, or in other digital data collections. Special tools like the ICD-11 Coding Tool facilitate finding specific ICD-11 codes for any of the several dimensions that define an ICD-11 entity or category. Additional details can be added using multiple codes for one condition. Retaining the unique identifier of the coded ICD-11 entity allows the same information to be reused across different translations. WHO has developed ICD web services (<https://icd.who.int/icdapi>); designed to support interoperable machine-to-machine interaction.

1.1.5.2 Use of ICD–11 in an analogue paper-based setting

ICD-11 can be used as an analogue printed version, if needed. Information is reported on hard copy documents and then coded manually with the ICD-11. It should be noted that paper-based recording requires manual transcription of the information into electronic systems and consideration should be given to moving to electronic reporting as early as possible in the information chain. Paper-based recording may cause problems with readability and timeliness of coded data. ICD-11 supports many ways of computer assisted coding including embedding of instructions for code combinations and other possible plausibility checks. The long-term goal for all users should be coding using ICD-11 in an electronic environment.

In the print version, the information is divided into 3 volumes, the Tabular list, the Reference Guide, and the Index. All three are needed to use the ICD correctly.

1.1.5.3 Electronic version

In the browser version of the ICD, most information is interlinked and visible in the relevant context. The WHO provides this version for browsing ICD-11 in multiple languages (linked from <https://icd.who.int>). This tool allows the user to retrieve concepts by searching for terms, anatomy or any other element of the content model. With the browser, users can also contribute to the updating and continuous improvement of ICD with comments and suggestions. Such input is reviewed for consideration for inclusion on an annual basis.

ICD–11 can also be accessed using web services with user specific software. The IT guide to the ICD provides more details on compatibility requirements can be found on the following page: <https://icd.who.int/icdapi>. Both the web services and the online browser allow access and searching of all Tabular lists of the ICD, including for mortality and morbidity statistics, primary care, or for specialty linearisations for certain specific domains.

1.1.6 Links with other Classifications and Terminologies

ICD coded entities or categories can be used in conjunction with other relevant health classifications and terminologies to fully document an episode of care, or a case for research.

1.1.6.1 Integrated use with Terminologies

Classification involves grouping information according to logical rules. Terminology allows the reporting of information at any desired level of detail: for example, body parts, findings,

or other elements that constitute a disease. Only items defined in a terminology can be reported (i.e. terminologies have no mechanism to report new information that has not previously been added to the terminology). In contrast, a classification has residual classes ('other specified' and 'unspecified') that ensure that all cases can always be classified. In a terminology, similar to a modern classification, a disease can be defined, for example by establishing linkages between its elements, such as anatomy or findings. Terminologies retain the information without emphasising any aspect of the recorded information.

In contrast, classifications allow for identification of 'relevant parts' of the content, for example, for public health purposes. International agreement about these relevant parts ensures sure that the aggregated information is internationally comparable. The standardised use of the aggregation logic of a classification and the standardised use of the detailed information of a terminology aim at the same result: comparability. Transparent international agreement processes are necessary in both cases, with integration of index and tabular lists, ICD has moved towards having its own terminology component.

Terminologies and classifications should be considered complementary. ICD-11 Foundation is the terminological part of ICD-11 derived from the former ICD-10 Index and other sources. Third party terminologies ideally are linked to the ICD-11 Foundation in 1:1 relationships or "signposted from ICD-11 foundation in 1:n relationships. These links shall follow the recommendations by the mapping paper.

1.1.6.2 Functioning in ICD and joint use with ICF overview

Historically, the ICD has used certain disability concepts as common disease or disorder entities, such as: blindness, deafness, learning disability, or paraplegia, as well as certain disability concepts for other purposes, such as 'disability as a sequela of injury', and 'limitation of activities due to disability'. The ICF was developed after the publication of ICD-10.

In ICD-11, links with ICF have been created in terms of aligning the concepts of disease and functioning and facilitating the joint use.

Conceptual alignment between ICD-11 and ICF has taken place in several areas.

- Signs and symptoms in the ICD are aligned with body functions in the ICF, and 'factors influencing health status' in the ICD align with contextual factors in the ICF.
- ICD-11 categories related to visual impairment in the Ophthalmology Chapter (Chapter 09) have been aligned with the respective ICF taxonomy and definitions for seeing functions.

Additional selected ICF categories are drawn from the component Activities and Participation and help to describe the functional limitations commonly associated with specific health conditions in a functioning pattern. The impact of a disease or disorder in the daily activities of a person may vary depending on the severity of the condition as well as the contextual factors (e.g. environmental factors) and possible co-morbidities. The ICD takes an approach that identifies 'severity' as a property of the disease/disorder and describes the impact of the health condition on the daily life of a person with a set of functioning codes.

An optional functioning section has been embedded in ICD-11 to enable the classification and measurement of the impact of health conditions in terms of functioning.

The ICD-11 functioning section provides two options for impact assessment and documentation:

- use of an ICF-based generic assessment instrument (i.e. World Health Organization Disability Assessment Schedule (WHODAS) 2.0) and the Model Disability Survey (MDS) to generate an overall and domain specific functioning score. In order to obtain such scores, WHODAS 2.0 (like other clinical rating scales) has to be used as a whole (i.e. the full battery of questions). Further instructions on the use of WHODAS 2.0 are provided in the WHODAS Manual.
- use of a short-list of functioning categories derived from ICF Annex 9. These categories should be used with a generic 5-point qualifier indicating severity for reporting of functioning in health information systems.

Overall, the linkage between ICD-11 and ICF may assist with the following use cases:

- evaluation for general medical practice (e.g. work in capacity assessment)
- evaluation for social benefits (e.g. disability pension)
- payment or reimbursement purposes
- needs assessments (e.g. for rehabilitation, occupational assistance, long-term care)
- outcome evaluation of interventions

Wherever full *functioning* reporting is desired and required, the ICF should be used. (see Section 2.11 for further information)

1.2 Structure and taxonomy of the ICD

The chapter and block structure of the ICD has evolved in 11 iterations of the classification over 150 years. The authoring of ICD follows a set of rules that ensure the functional and structural integrity of the classification. The evolution of the ICD carefully balances the need for categories that match current knowledge while allowing statistical comparability over space and time.

The chapter structure of ICD reflects major aspects of diseases. Chapters are not intended to delimit areas of medical expertise or domains of specialties. The link to any specialty or reimbursement schemes is secondary. The ICD has categories for diseases, disorders, syndromes, signs, symptoms, findings, injuries, external causes of morbidity and mortality, factors influencing health status, reasons for encounter of the health system, and traditional medicine. ICD-11 complements these by allowing capture of additional details such as anatomy, substances, medications, medical devices, infectious agents, place of injury, and many more. ICD-11 also comes with a set of rules and explanations for its use and necessary metadata. Required reporting formats are also included.

The most widespread use of ICD over time and geographically is for cause of death (mortality) statistics. ICD is also used for classification of clinical documentation, to provide standardised, language independent information for morbidity use, such as resource allocation, casemix, patient safety and quality of care, as well as primary care and research. ICD and its descriptions are also used as a framework in legislation.

1.2.1 Taxonomy

A statistical classification of diseases must be confined to a limited number of mutually exclusive categories able to encompass the complete range of morbid conditions. The categories are chosen to facilitate the statistical study of disease phenomena. Every disease or morbid condition must have a well-defined place in the list of categories. Consequently, throughout the classification, there are residual categories for other and miscellaneous conditions that do not have their own unique category or which cannot be allocated to the more specific categories. However, the ICD-11 Uniform Resource Identifiers (URI) allow retention of such detail for future analysis. The following measures apply in determining whether an entity qualifies to become a unique category:

1. Epidemiological evidence: frequency analyses of coded mortality and morbidity data
2. Clinical evidence: disease evidence provided by the medical specialties
3. Granularity: minimum detail reported and useful in mortality or primary care
4. Continuity: preservation of the level of detail pre-existing in ICD
5. Parsimony: the need to limit the number of categories for international mandatory reporting

A statistical classification can allow for different levels of detail if it has a hierarchical structure and subdivisions. A statistical classification of diseases should retain the ability to both identify specific disease entities and to allow statistical presentation of data for broader groups, thus enabling the attainment of useful and understandable information. The same general principles apply to the classification of other health problems and reasons for contact with health care services, which are also incorporated in the ICD. The ICD was developed as a practical, rather than a purely theoretical classification, in which there are a number of compromises between classification based on aetiology, anatomical site, circumstances of onset, or other criteria.

ICD-11 draws extensively on the method of combining several codes to describe a clinical condition to the desired level of detail. Its electronic architecture allows assignment of unique identifiers to any condition listed - independently whether the condition is grouped in a statistical class or whether it represents a class of its own. The two approaches together allow the option of keeping coding simple where diagnostic detail is limited; and alternatively adding detail where diagnostic reporting requires a high level of comprehensiveness and sophistication.

1.2.1.1 Content model and definition of disease

Content model

For further information please see [3.4 The Content Model](#).

Definition of disease

A disease is usually defined using a set of relevant aspects drawn from the pattern below. A disease is a set of dysfunctions in any body system defined by:

Property	Description
Symptomatology or manifestations	Known pattern of signs, symptoms, and related findings
Aetiology	An underlying explanatory mechanism
Course and outcome	A distinct pattern of development over time
Treatment response	A known pattern of response to interventions
Linkage to genetic factors	E.g. genotypes, patterns of gene expression, etc.
Linkage to environmental factors	Factors of the environment that trigger presence of a disease

1.2.2 ICD Chapter structure

The ICD is a variable-axis classification. The structure was developed out of that proposed by William Farr in the early days of international discussions on classification structure: epidemic diseases, constitutional or general diseases, local diseases arranged by site, developmental diseases, injuries.

These groups remain in the chapters of ICD–11. The structure has stood the test of time and, though in some ways arbitrary, is still regarded as more useful for general epidemiological purposes than any of the alternatives tested. The conservation of the structure acknowledges the need for stability while allowing incorporation of additional sections.

Special groups bring together conditions that would be inconveniently arranged for epidemiological study if they were to be scattered, for instance in a classification arranged primarily by anatomical site. **These conditions formulate the ‘special groups’ chapters:**

Chapter	Title
1	Certain infectious or parasitic diseases
2	Neoplasms
3	Diseases of the blood or blood-forming organs
4	Diseases of the immune system
18	Pregnancy, childbirth, or the puerperium
19	Certain conditions originating in the perinatal period
20	Developmental anomalies
22	Injury, poisoning or certain other consequences of external cause

The distinction between the ‘special groups’ chapters and the ‘body systems’ chapters has practical implications for understanding the structure of the classification, for coding to it, and for interpreting statistics based on it. It has to be remembered that, in general, conditions are primarily classified to one of the ‘special groups’ chapters.

Where there is any doubt as to where a condition should be positioned, the ‘special groups’ chapters take priority. This principle is enforced in the ‘excludes’ notes at the beginning of each chapter in the ICD.

1.2.3 Guiding principles for classification of special concepts

1. Clinical findings are located in Chapter 21 'Symptoms, signs or clinical findings, not elsewhere classified'. (e.g. 'Abnormal serum enzyme levels' or 'Results of function studies of the circulatory system')
2. Manifestations of diseases, a relevant point for health intervention, are 'clinical manifestations' located in the body system chapter where they manifest. The underlying condition has to be coded as well. (e.g. myocarditis)
3. Syndromes, where the aetiology is unknown, are allocated with the most relevant body system. (e.g. Costen syndrome is in the 'Digestive' chapter)
4. The number of categories with 'due to' in the title are restricted to certain exceptions. (e.g. acute bronchiolitis due to respiratory syncytial virus)
5. Very specific, chronic, postprocedural conditions are grouped at the end of the body system chapter where they manifest. (e.g. lymphoedema due to surgery or radiotherapy). Residual categories do not exist for these groups.
6. Acute postprocedural complications are identified by combinations of codes from body system or injury chapters, and external causes chapter (e.g. an unintentional puncture of an organ during an intervention is classified with a code for the injured organ (harm), a code describing what surgery caused the injury (cause), and a code identifying the unintentional puncture as the mode/mechanism of injury.).
7. Categories with mention of 'multiple' are restricted to exceptions and require coding of the different multiple conditions individually (e.g. multiple injuries are to be coded individually when possible).
8. Categories with mention of 'sequelae' are restricted to exceptions. The specific condition resulting as a sequela needs to be coded along with the underlying cause. In some instances, they will continue to exist with the label 'late effects of...' (e.g. late effects of cerebrovascular disease or late syphilis). 'Sequelae' include residual effects of diseases or disorders, injuries or poisonings specified as such, or as late effect of, arrested, cured, healed, inactive, old or quiescent condition unless there is evidence of active disease.
9. Categories with mention of 'history of' are limited to exceptions (e.g. personal history of malignant neoplasms lists only the more frequent anatomical sites).
10. High level groupings need to be meaningful.
11. Residual categories exist only where they are meaningful. (e.g. where conditions are either congenital or acquired, there is no 'other' residual, but there will be an 'unspecified' option)

1.2.4 General features of ICD-11

The main structural innovation of ICD–11 is that it is built on a Foundation Component from which the Tabular lists (such as the classification for morbidity and mortality statistics) are derived.

1.2.4.1 Code structure

The codes of the ICD–11 are alphanumeric and cover the range from 1A00.00 to ZZ9Z.ZZ. These are referred to as stem codes. The structure of stem codes is described below:

- The first character of the code always relates to the chapter number. It may be a number or a letter.
- Codes starting with 'X' indicate an extension code (see [2.9](#)).
- There is always a letter in the second position to differentiate ICD-11 codes from the codes in ICD-10.
- The inclusion of a forced number at the third character position prevents spelling 'undesirable words'.
- The letters 'O' and 'I' are omitted to prevent confusion with the numbers '0' and '1'.

For example, [1A00](#) is a code in Chapter 01, and [BA00](#) is a code in Chapter 11.

For example: ED1E.EE

- E corresponds to a 'base 34 number' (0-9 and A-Z; excluding O, I);
- D corresponds to 'base 24 number' (A-Z; excluding O, I); and
- 1 corresponds to the 'base 10 integers' (0-9)
- The first E starts with '1' and is allocated for the chapter. (i.e. 1 is for the first chapter, 2: chapter 02, ... A chapter 10, etc.)

The terminal letter Y is reserved for the residual category 'other specified' and the terminal letter 'Z' is reserved for the residual category 'unspecified'. For the chapters that have more than 240 blocks, 'F' ('other specified') and 'G' ('unspecified') are also used to indicate residual categories (due to limitations in the coding space).

Throughout the Reference Guide a dash indicates that more detailed codes exist, and should be used as appropriate. For example

- [8B24](#) - covers codes from [8B24.0](#) to [8B24.Z](#)
- [CA23.0](#) - covers [CA23.01](#) to [CA23.02](#)

1.2.4.2 Uniform resource identifiers

All entities have a Uniform Resource Identifier (URI) which is a string of characters that uniquely identifies a particular ICD-11 entity. Each entity has a specific place in the hierarchy of groups and categories.

1.2.4.3 Block codes

Higher level entities in ICD-11 (called 'blocks') may be used for reporting aggregated statistics. However, blocks do not have category codes as they are not supposed to be used in coding. Blocks have their own URIs (e.g. the URI for Neoplasms is <http://id.who.int/icd/entity/1630407678>). Blocks may also be referred to by block IDs. The code structure for block IDs are 11 characters long (e.g. "BlockL1-1A0").

1.2.4.4 Stem codes

Codes in the Tabular list of ICD-11 that can be used alone. Stem codes may be entities or groupings of high relevance in any of the use cases, or clinical conditions that should always be described as one single category. Stem codes are designed to ensure that in use cases that require only one code per case, meaningful information is collected.

1.2.4.5 Extension codes and postcoordination

The extension codes are comprised of groups of codes e.g. anatomy, agent, histopathology and other aspects that may be used to add detail to a stem code. Extension codes are not to be used alone in the context of statistical classification but must be added to a stem code. Extension codes may be used in another context, e.g. for device documentation. Not all extension codes can be used with every stem code. Refer to [2.9](#).

Postcoordination is a notable new feature in ICD-11 that creates the ability to link core diagnostic concepts (i.e. stem+stem code concepts) when desired, and/or to add clinical concepts captured in extension codes to primary stem code concepts. The linked diagnostic concepts are called a cluster. It should be emphasised that the postcoordination ability inherent in ICD-11 is one of the significant changes compared with ICD-10. A cluster describes one clinical concept as described by the health care practitioner. Postcoordination is by matter of principle permitted with all axes, as long as at least one stem code is used. API, Browser and coding tool facilitate postcoordination in certain areas. Other areas of postcoordination have to be coded individually, but may be supported by future updates of ICD-11, where need arises.

1.2.4.6 Other general features

- ICD-11 categories have **short descriptions**, plus long descriptions labelled ‘additional information’. The short description is a maximum of 100 words relating to the entity that states things that are always true about a disease or condition which are necessary to understand the scope of the rubric. It appears in the Tabular list of the classification. The long ‘additional information’ is the full description, without length restriction.
- **Special tabulation lists** (refer to section [2.25](#)) continue to exist in ICD-11, but there are three additional lists - the Startup Mortality List (SMoL), the list for verbal autopsy, and the list for infectious diseases by agent. Specialty linearisations allow the representation of content from the perspective of a specialty, such as dermatology or neurology, through the creation of subsets, and through the precoordination of more detail, if desired.

1.2.5 Foundation Component and Tabular lists of ICD-11

The Foundation Component is a multidimensional collection of all ICD entities. It is an underlying database that holds all necessary information to generate print versions of the tabular list and the alphabetical index, as well as additional information that is needed to generate specialty linearisations of ICD-11 and country-specific modifications. Entities can be diseases, disorders, injuries, external causes, signs and symptoms. Some entities may be very broad, for example ‘injury of the arm’, while others are more detailed, for example ‘laceration of the skin of the thumb’. The Foundation Component also has the necessary information to use the entities to build a tabular list. The Foundation Component includes information on where and how a certain entity is represented in a Tabular list, whether it becomes a grouping, a category with a stem code, or whether it is mentioned as an index term in a particular category.

Several different Tabular lists can be built from the Foundation Component. Drawing on the same Foundation Component, a set of tabular lists that builds on the same hierarchical tree structure can be created – producing congruent tabular lists. The Foundation component includes instructions on how to combine certain codes in a Tabular list to achieve more

detail in coding. These rules help coders and computer systems to visualise the permitted code combinations when they are using a Tabular list.

In a Tabular list, entities of the Foundation Component become categories. The categories are mutually exclusive and jointly exhaustive and linked to a mono hierarchical tree (that is, they have only one parent). The information related to an entity that has become a category and has multiple parents is still available from the Foundation Component. This information can be used to visualise that category in more than one place in the Tabular list, e.g. by showing them in black font in its place for reference tabulation and in grey font in any other place for browsing or alternative tabulations. ICD–11 has multiple congruent tabular lists with varying levels of detail.

1.2.5.1 Precoordination and Postcoordination in ICD-11

A health condition may be described to any level of detail, by applying more than one code, or by ‘postcoordinating’ (i.e. combining):

- two or more stem codes, (e.g. code1/code2)
- stem codes with one or more extension codes. (e.g. stem code&extension code1&extension code2)

In this manner, the classification can address a large number of clinical concepts with a limited range of categories.

Stem codes contain all pertinent information in a pre-combined fashion. This is referred to as ‘precoordination’. When additional detail that pertains to a condition is described by combining multiple codes, this is referred to as ‘postcoordination’. The combination of codes is called a cluster. Refer to [2.10](#).

1.2.5.2 Multiple parenting

An entity may be correctly classified in two different places, e.g. by site or by aetiology. For a disease, such as oesophageal cancer, this would mean that it could be classified to cancers (malignant neoplasms) or to conditions of the digestive system. In the same way, cerebral ischaemic conditions could be classified to the vascular system or to the nervous system. A decision about which place a condition is in depends on international agreement and legacy.

1.2.6 Language independent ICD entities

ICD-11 entities are language independent. The maintenance of the ICD-11 on an international level is handled in the English language but the content model of ICD–11 is language independent and allows binding of any desired language to the elements of its Foundation Component. In this way, an international translation base facilitates translations or multilingual browsing. (See [3.12 Annex A: ICD-11 Updating and Maintenance](#))

1.3 Main uses of the ICD: Mortality

Mortality statistics are widely used for medical research, monitoring of public health, evaluating health interventions, and planning and follow-up of health care. Rules adopted by the World Health Assembly (WHA) regarding the selection of a single cause or condition, from death certificates, for routine tabulation of mortality statistics are provided to

standardise production of mortality data. Implementation of the ICD for mortality requires establishment of an infrastructure for reporting and storing information, design of information flows, quality assurance and feedback, and training for classification users working with the input or output of data.

1.3.1 What is coded: Causes of death

The description of a single underlying cause of death, and selected approaches to capture further information on causes of death also reported on a death certificate, enables the identification of trends in health for a given population.

Effective public health interventions prevent harm or death by breaking the chain of events that lead to harm. For this purpose, the underlying cause of death has been defined as ‘(a) the disease or injury which initiated the train of morbid events leading directly to death, or (b) the circumstances of the accident or violence which produced the fatal injury’, and is selected for routine single-cause tabulation of mortality statistics. (See Section [2.21](#) for more information.)

1.4 Main uses of the ICD: Morbidity

Morbidity data are used for statistical reporting mostly at national or local levels. While some of this statistical reporting is conducted within an academic research context, it is commonly conducted in applied settings to inform health system and public health agency decision-making. ICD coded data also forms the basis for different casemix systems, such as different varieties of Diagnosis Related Groups (DRGs). Coded morbidity data can also be used to inform a variety of clinical guidelines through provision of Foundation Component information on burden of disease.

1.4.1 What is coded: patient conditions

The definition of main condition relates to the description of an episode of hospital-based care. The health care practitioner should record and identify as the main condition the one condition that is determined to be the reason for admission, established at the end of the episode of health care.

The health care practitioner responsible for the patient’s treatment is also responsible for documenting the patient’s health conditions. This information should be organised systematically by using standard recording methods. A properly completed record is essential for good patient management. It is also an essential prerequisite to the creation of a valid coded record of patient diagnoses, derived through a coding process from written information describing the patient’s medical condition. When a sound written record of patient conditions is available, successful coding of this information in ICD and associated classifications produces a valuable source of epidemiological and other statistical data on morbidity and other health care problems. The person transforming the information on the stated condition to codes (the ‘coder’) may be the health care practitioner or a clinical coder (who is not responsible for the patient’s treatment). In the latter situation, which is common among member countries, the coder depends on the adequacy of clinical documentation of patient conditions by health care practitioners in the medical record. The importance of clinical documentation by health care practitioners as the starting point for coded health data cannot be overstated, and needs to be underlined as being a matter of key significance

within countries and internationally. This has implications for education on health information and clinical documentation within health care practitioner training programs. (See Section [1.4](#) for more information.)

1.5 Traditional Medicine

Traditional Medicine (TM) is an integral part of health services provided in many countries. International standardisation by including Traditional Medicine within the ICD allows for measuring, counting, comparing, formulating questions and monitoring over time. Although some countries have had national Traditional Medicine classification systems for many years, information from such systems has not been standardised or available globally.

It is recommended that coding of cases with ICD-11's chapter on Traditional Medicine disorders and patterns (TM1) be used in conjunction with the Western Medicine concepts of ICD Chapters 1-25.

As with other ICD chapters, the TM1 chapter is not designed to assess TM practice or the efficacy of any TM intervention. However, as a tool for classifying, diagnosing, counting, communicating and comparing TM conditions, it assists research and evaluation to assess the safety and efficacy of TM.

See [3.12 Annex A: ICD-11 Updating and Maintenance](#) for more detailed information.

1.6 ICD maintenance

The ICD maintenance process allows for the updating of the ICD following evolution in the understanding of diseases, treatments, and prevention. It also ensures improvements and clarifications coming from daily use of ICD, and requests by Member States or any other interested party.

A standardised open process has been established to ensure that the proposed updates are collected, routed, reviewed, and duly considered before being implemented. A proposal and review mechanism on an online platform makes the process transparent. Workflows ensure that proposed changes are considered both from a medical and scientific perspective and from their value and place in a particular use case (See [3.12 Annex A: ICD-11 Updating and Maintenance](#)).

1.6.1 Guiding principles of authoring process

Allocation of entities in the classification follows a set of rules that serve to maintain the structural and functional integrity of the classification. The core set of rules listed here is complemented by additional rules that address special cases or serve to ensure consistent user guidance. They are listed in order of priority.

1. No changes to the classification, including movement of categories or groups between chapters, without rationale and documented change in aetiology or prevention method. (e.g. Chapter 04 - 'Diseases of the immune system' was added as a new chapter as there was sufficient scientific evidence to support this move). Alternatively, it was suggested to move 'wounds of skin' to 'Diseases of the skin'. The

wound of the skin, being an injury, remains grouped with injuries because prevention will focus on the cause of the wound.

2. Conditions are classified predominantly by their aetiology.
 - Local manifestations of important ‘aetiologies’ are located in the aetiology chapter (e.g. Viral hepatitis is in ‘Certain infectious or parasitic disease’).
 - Where one condition can be due to multiple different aetiologies, and it is more relevant to retain the affected body system, it is usually classified with the body system (e.g. some gastric ulcers are caused by bacteria, but they remain in the ‘Digestive system’ chapter).
 - Where the aetiology of the condition is unknown, it is allocated to the most relevant organ system (e.g. Costen syndrome is in the ‘Digestive system’ chapter).
 - Systemic ‘aetiologies’ are primarily in their relevant aetiology chapter (e.g. Idiopathic inflammatory myopathy is in “Diseases of the immune system”).
3. Conditions that could arguably be in two or more places of the classification remain in their legacy location.
 - For example, injuries of the eye are equally important for the eye and their prevention. Despite the suggestion of including them in the eye chapter, they remained where they were, in the injury chapter.
 - Where aetiology and body system are equally important, the legacy location remains unchanged (e.g. ocular motor nerve palsies).
4. Keeping a group of subtypes together in one location may override anatomical or aetiological considerations (e.g. human prion diseases - some have a genetic component, others a transmissible component).

1.6.2 Improving user guidance

The following rules serve to provide user guidance. Users may expect to find conditions in certain places when browsing the tree structure. User groups may need to group data or create subsets for other reasons. The multiple parenting in the Foundation Component serves to address that issue.

1. Where a condition could be in two or more places, identify these other places and add them as secondary parents, e.g. malignant neoplasm of the colon is coded to the neoplasm chapter, but is also shown in the chapter of diseases of the digestive system. In case a set of conditions needs to be shown in more than one place and there is no grouping matching that set, create a window (no primary children, no terms, no residual categories) in the appropriate place.
2. Where a condition could be confused with another condition bearing a similar name, add an exclusion note. (e.g. ‘Influenza due to seasonal influenza virus’ has a note ‘Exclusion: Haemophilus influenzae [H. influenzae] meningitis’).
3. Where alternative ways of tabulating data are required, create a special linearisation list as a second parent (e.g. infectious diseases by agent). The coding scheme of the individual entries will remain the one used for the full international classification.
4. Where diseases of certain body systems are spread across different chapters, allow for a specialty linearisation of the pertinent diseases. The coding scheme of the

individual entries will remain the one used for the full international classification. Currently there are specialty linearisations for primary care, dermatology, neurology, ophthalmology, and in special cases such as the International Classification of Disease for Oncology (ICD-O) and the International Classification of External Causes of Injury (ICECI).

5. **Deprecated term:** A term/concept that has been deemed invalid (outdated) for inclusion in the current version of ICD-11, which may still be recorded. A user who performs a search using a term that has been deemed “deprecated” will see (deprecated) displayed following the invalid (outdated) term. “Deprecated” terms are hidden from the ICD-11 browser (tabular) and do not appear under the “All Index Terms” heading.

1.6.3 Introduction to the ICD-11 Update Process

The ICD-11 classification updating is carried out at different levels with different frequencies.

Continuous updates are made to the:

- ICD-11 Foundation Component
- ICD-11 index entries

Official releases of the ICD-11 classification are produced: - Annually for international use in mortality and morbidity (known as the ‘blue browser’).

- Every five years for updates that impact on the four- and five-character codes
- Every ten years for updates to the mortality and morbidity rules that have significant impact on statistical output and need to be updated in long-term cycles

A standardised process has been established to ensure that the proposed updates are collected, routed, reviewed, and duly considered before being implemented. Any individual user of the classification can submit a proposal for an update to the ICD. Such updates can refer to one or more entities of the ICD. They may address the position of entities in a tabular list, in the Foundation Component, and any element of the content model.

The maintenance platform of ICD-11 (known as the ‘orange browser’) is used for proposals and comments. Any input to ICD-11 and its components requires proper referencing of sources, details of scientific evidence, and permission from the owner of any copyright materials (where applicable).

1.6.4 National Modifications for morbidity coding

The use of ICD in the specific context of the health care system of a country may require detail that is not currently part of ICD-11, for example, due to specific settings or due to reimbursement system requirements. Such changes will be subject to the same international process as are all other changes to ICD and will then become part of the Foundation Component and eventually of the Mortality and Morbidity Statistics (MMS), ideally prior to their implementation in the requesting country.

A situation may arise, where a national government or a related national institution needs a modification to be implemented immediately. In such circumstances, conflicts with the

current Foundation Component must be avoided, and the relevant changes will be subject to special mechanisms for the international updating process. All countries planning to produce national modifications must make relevant contractual arrangements with WHO. This includes regulations on distribution within and outside the respective country and the resources necessary for the WHO to add such changes to the foundation.

For developing a national modification of ICD-11 the following rules must be adhered to:

1. Ideally, modifications will be agreed by the ICD-11 maintenance bodies before they are implemented nationally.
2. Modifications should not impact on morbidity and mortality statistics, and should not conflict with the foundation.
3. Approval of all national modifications will be subject to consideration of whether suitable additional detail already exists in the foundation.
4. If a change is made to the international version of ICD-11 the respective national modification must incorporate the change as soon as possible.

Example

‘Diabetes Type 1’ in the WHO version of ICD-11 is classified to [5A10](#). A national modification may require additional detail to be added to the ICD-11 codes. For example, ‘Diabetes Type 1, uncontrolled’ could be added as a subcategory to [5A10](#), as 5A10.1 Diabetes Type 1, uncontrolled. However, when the mechanism of postcoordination provides the necessary detail, the addition of a new subcategory may not be necessary.

2 Part 2 - Using ICD-11

2.1 Basic coding and reporting guidelines

Coding is the assignment of one or more codes in order to represent the meaning of a condition in as much detail as required. Before attempting to code, the coder should be acquainted with the principles of classification and coding. In some instances, using one code will provide sufficient detail. In other instances, it may be necessary to use several codes together to express the level of detail required by the use case, setting, or laws.

For coding, it is recommended to use the ICD-11 Smart coding tool that can be used online and offline. It provides users with a simple automated way to find and select the needed categories. When the search shows a cluster of codes, rather than a single stem code, the tool can return the assembled cluster.

Software must not include lists or other prompts to guide the recording or coding, as these necessarily limit the range of diagnoses and therefore have an adverse effect on the accuracy and usefulness of the record and report.

There may be an alternative way, to use the print version of ICD-11 but this option is not recommended as its time consuming and may cause confusion in finding and selecting the correct codes or code combinations without the hints that the electronic coding tool provide. In case no electronic environment is available, specific solution can be sought in collaboration with WHO.

2.2 Tabular List, Special Tabulation Lists, Qualifiers, and Modifiers

The Tabular list is an alphanumeric listing of diseases and disease groups, inclusion and exclusion notes, and some coding rules. Chapters 1 to 25 of the ICD contain approximately 15 000 entities at the four-, five- or six-character level.

In addition, there is a section on extension codes ([2.9](#)) and one on Traditional medicine ([1.5](#)). The Special tabulation lists are presented at the end of the Tabular list. Special tabulation lists are not designed for coding but are for tabulation and reporting only ([2.25](#)).

2.3 Index

The Alphabetical Index is a list of more than 120 000 clinical terms (including synonyms or phrases). The index is used to find the relevant ICD codes or code combinations for clinical terms. The mention of a term in the index exclusively serves coding. Mention of a term in the index does not mean approval or endorsement of a particular condition.

2.4 Reference Guide

The Reference Guide contains an introduction to the context, components, and intended use of the ICD. It describes the diverse components of ICD-11, provides guidance for certification, recording, rules for mortality coding (i.e. causes of death statistics) and morbidity coding (e.g. hospital statistics) and lists for tabulation of statistical data.

2.5 Browser and coding tool

The WHO provides a browser and coding tool ('blue browser') for ICD-11 in multiple languages <https://icd.who.int>. This tool allows the user to retrieve concepts by searching for terms, anatomy or any other element found within the content of ICD-11 (see [1.2.5 Foundation Component and Tabular lists of ICD-11](#)).

WHO also provides a maintenance platform ('orange browser'), (see [3.12 Annex A: ICD-11 Updating and Maintenance](#)). This tool allows the users to contribute to the updating and continuous improvement of ICD-11. Such input is submitted in the form of a proposal with a detailed rationale, scientific evidence and peer reviewed references to justify consideration for inclusion on an annual basis (see [3.12 Annex A: ICD-11 Updating and Maintenance](#)).

For coding, the WHO provides the ICD-11 Smart coding tool, a simple automated way to find and select the needed categories.

ICD-11 can also be accessed using web services with user-specific software. The IT guide to the ICD provides more details on compatibility requirements.

Both the web services and the online browser allow access to all Tabular lists of the ICD, for mortality and morbidity statistics, primary care, or for a specialty linearisation for use in certain specialised domains.

2.6 Coding step by step – clinical term

The table below compares the coding steps in a paper and an electronic environment. The essential component of coding is finding a match to the reported clinical term – having a good dictionary in the relevant language, and verifying the resulting code against additional